



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE NATIONAL INSTITUTE OF ENGINEERING
(An Autonomous Institution)



Project Synopsis

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USN	Name	Email-ID	Mobile Number
4NI23CS133	P Rishika	2023cs_prishika_c@nie.ac.in	7619543704
4NI23CS140	Poojitha M S	2023cs_poojithams_c@nie.ac.in	6360534181
4NI23CS157	Priyanka Pramod Daivagna	2023cs_priyankapramoddaivagna_c@nie.ac.in	9972679634
4NI23CS187	Sanjana S Shetty	2023cs_sanjanasshetty_c@nie.ac.in	9972197827

Project Title: An Explainable and Robust NLP Framework for Clickbait Detection Against Adversarial Attacks

Introduction: The rapid growth of online news and social media has significantly increased the spread of clickbait content. Clickbait headlines are intentionally designed to attract users' attention by creating curiosity or exaggerating information, often leading to misleading or low-quality content. Such headlines contribute to misinformation, reduce user trust, and negatively affect the online information ecosystem. Recent advances in Natural Language Processing (NLP), particularly transformer-based models such as BERT and RoBERTa, have achieved high accuracy in clickbait detection. However, these models are vulnerable to adversarial attacks, where attackers introduce minor character-level modifications such as replacing letters with visually similar symbols (e.g., $o \rightarrow 0$, $a \rightarrow @$, $e \rightarrow 3$). Although these changes are almost unnoticeable to humans, they can significantly reduce the performance of NLP classifiers. The proposed project develops a robust defensive framework that not only detects clickbait but also identifies adversarial manipulations, restores the original text, provides explainable predictions using Explainable AI (XAI), and continuously improves itself by learning from newly observed attack patterns.

Objectives:

- To develop a robust clickbait detection system using transformer-based models.
- To detect adversarial manipulations in clickbait headlines.
- To restore adversarially modified text before classification.
- To improve classification accuracy under adversarial conditions.
- To provide interpretable predictions using Explainable AI techniques.
- To implement continuous learning for adapting to evolving attack patterns.
- To evaluate the framework using standard benchmark datasets.

Software Requirements –

Operating System

- Windows 11 / Ubuntu Linux

Programming Language

- Python 3.11+

Libraries & Frameworks

- PyTorch
- Hugging Face Transformers
- Scikit-learn
- Pandas
- NumPy
- NLTK
- spaCy
- RapidFuzz / SymSpell
- LIME
- SHAP
- Flask / FastAPI
- Streamlit

Development Tools

- Visual Studio Code
- Google Colab
- Git & GitHub
- Jupyter Notebook

Hardware Requirements –

- **Processor:** Intel Core i3/i5 or AMD equivalent
- **RAM:** Minimum 4 GB (8 GB recommended)
- **Storage:** Minimum 100 GB free disk space
- **System Architecture:** 64-bit Computer
- **Display:** 1366 × 768 resolution or higher
- **Network:** Stable Internet connection (for downloading datasets, libraries, and model updates)

Abstract: Clickbait headlines are widely used to attract user attention and increase website traffic, but they often contribute to misinformation and poor user experience. Although transformer-based NLP models have significantly improved clickbait detection, they remain vulnerable to adversarial attacks involving small character-level perturbations. Such attacks can bypass detection systems while preserving human readability. This project proposes a robust and explainable clickbait detection framework capable of defending against adversarial text manipulations. The framework consists of five major components: adversarial attack detection, text restoration, robust clickbait classification using RoBERTa/BERT, explainable prediction using LIME and SHAP, and continuous learning through periodic retraining. Before classification, manipulated headlines are detected and restored to their original form, ensuring that the classifier

receives clean input. Explainability techniques provide transparency by highlighting influential words responsible for predictions, while continuous learning enables the system to adapt to emerging attack strategies. The framework is evaluated using the Webis Clickbait Corpus 2017 and adversarially generated datasets. Experimental results are expected to demonstrate improved robustness, higher detection accuracy, enhanced interpretability, and better resilience against evolving adversarial attacks compared to conventional clickbait detection systems.

Literature Survey :

Sl.no	Paper Title	Authors	Methodology	Limitation
1.	Prompt-Tuning for Clickbait Detection via Text Summarization	Haoxiang Deng, Yi Zhu, Ye Wang, Jipeng Qiang, Yunhao Yuan, Yun Li, Runmei Zhang	Uses prompt tuning with text summarization for clickbait detection.	No adversarial attack handling.
2.	Local Explainability-Based Model for Clickbait Spoiler Generation	Itishree Panda, Jyoti Prakash Singh, Gayadhar Pradhan	Applies Explainable AI for spoiler generation and prediction explanation.	No robustness against adversarial attacks.
3.	Deep Learning and Sentence Embeddings for Detection of Clickbait News from Online Content	Amara Muqadas, Hikmat Ullah Khan, Muhammad Ramzan, Anam Naz, Tariq Alsahfi, Ali Daud	Uses deep learning with sentence embeddings for clickbait classification.	Evaluated only on clean data.
4.	Clickbait Detection in News Headlines Using RoBERTa-Large Language Model and Deep Embeddings	Fawaz Khaled Alarfaj, Amara Muqadas, Hikmat Ullah Khan, Anam	Employs RoBERTa-Large with deep embeddings for classification.	No explainability or adversarial defense.
5.	Hard Label Adversarial Attack with High Query Efficiency Against NLP Models	Shilin Qiu, Qihe Liu, Shijie Zhou, Min Gou, Yi Zeng, Zhun Zhang, Zhewei Wu	Generates efficient hard-label adversarial attacks on NLP models.	Focuses only on attacks, not defense.

Existing System & its Drawback:

Existing System:Current clickbait detection systems primarily rely on transformer-based deep learning models such as BERT, RoBERTa, DistilBERT, XLNet, and CNN-based classifiers. These models perform well on clean datasets and accurately distinguish clickbait from non-clickbait headlines. Some recent approaches incorporate Explainable AI techniques to improve prediction transparency.

Drawbacks:

- Highly vulnerable to character-level adversarial attacks.
- Unable to detect manipulated text.
- No mechanism to restore corrupted headlines.
- Performance degrades significantly under adversarial conditions.
- Explainability and robustness are rarely integrated.
- Lack of continuous learning to adapt to new attack strategies.
- Focus only on classification rather than end-to-end defense.

Proposed System & Advantages:

Proposed System: The proposed framework introduces a comprehensive defensive pipeline for robust clickbait detection. Incoming headlines first undergo adversarial attack detection using rule-based and machine learning techniques. Manipulated text is then restored through normalization and fuzzy matching before being classified using a fine-tuned RoBERTa/BERT model. Explainable AI methods such as LIME and SHAP provide interpretable predictions, while newly encountered adversarial examples are stored and periodically used to retrain the model, enabling continuous learning.

Advantages:

- Detects adversarial manipulations before classification.
- Restores modified text to its original form.
- Maintains high accuracy under adversarial conditions.
- Provides transparent and explainable predictions.
- Continuously learns from newly observed attacks.
- Modular architecture allows future upgrades.
- Suitable for deployment as a REST API or web application.
- More robust than existing clickbait detection systems.
- Helps combat misinformation by improving the reliability of automated content moderation.

Guide Details:

Name : Dr Padmini M S

Designation : Assistant professor, Department of CS&E, NIE Mysuru

Mobile Number : 9739317964

College Address : No 50, Koorgalli Village, Hootagalli Industrial Area, Next to BEML, Mysore, Karnataka, 570018